Electronic Supplementary Material

Ionic liquids on uncharged and charged surfaces: *In situ* microstructures and nanofriction

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The thicknesses of the assemble IL layers on mica/HOPG surfaces (t_L) were determined by peak-to-valley distances (Fig. S1). In the determination of the layering thickness, the inhomogeneous bulky IL droplets on the solid substrates were excluded, because the friction coefficient was found to increase monotonically as the layering thickness (t_L) decreased, no matter what the thickness of the bulk IL (t_B) is. In our previous work [S1], we approximated the difference in the approaching atomic force microscopy (AFM) force–distance curve between the jump-in position of the piezo and the zero-force point as the apparent thickness of the IL film (t_A) on surfaces using AFM glass colloidal probes [2]. The measured apparent thickness (t_A) consists of three regions: 1) the IL adjacent to the solid substrate arranging in ordered layers (dense layer, t_{LD}); 2) the upper IL with decayed ordering into layers at a greater distance from the underlying solid surface (loose layer, t_{LL}); 3) the IL arranging further away from the substrate, contributing to the bulk phase (t_B). Noted that the AFM morphology determined layering thickness of the extended IL, t_L, is composed of both the dense (t_{LD}) and loose (t_{LL}) IL layers.

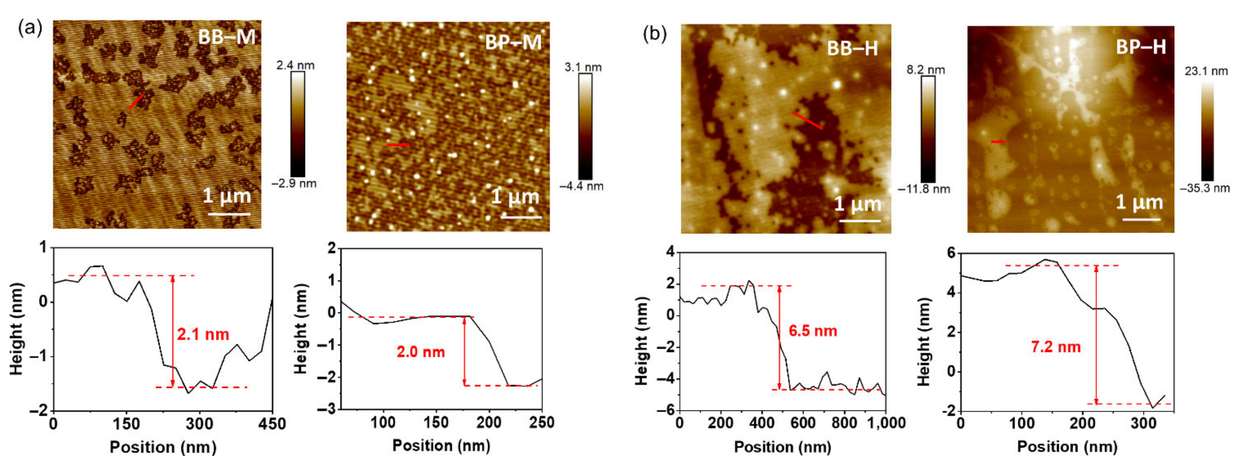

Fig. S1 AFM topographical images and corresponding cross-sectional profiles of the ILs (BB and BP) on surfaces of (a) mica (M) and (b) HOPG (H). Reproduced with permission from Ref. [S1], © the Owner Societies 2020.

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In this study, PMMA and SiO_2 probes were employed to check the influence of probe material on the *in situ* nanofrictional–microstructural changes of ILs at uncharged surfaces, so we showed the friction data on mica-supported ILs probed by PMMA and SiO_2 in Fig. 3(a). To further investigate the contribution of substrates to the *in situ* nanofrictional–microstructural changes of ILs at uncharged surfaces, we probed HOPG-supported IL systems by SiO_2 colloid probes (Fig. 3(b)) to compare with mica-supported IL systems by also SiO_2 probes (Fig. 3(a)). Thus, we did not focus on the friction data of HOPG-supported IL systems with PMMA probes, which were only shown in Fig. S2.

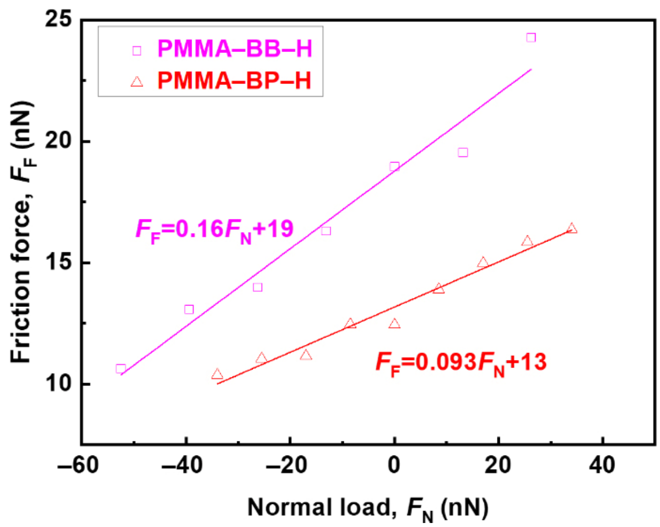


Fig. S2 Friction force (F_F) vs. normal load (F_N) for PMMA colloid probes sliding over the ILs (BB and BP) films at HOPG substrates.

It is noteworthy that the friction measurements in this study occurred between the probe and IL surfaces. In order to confirm that an IL film exists between the AFM probe and the substrate during friction measurements, we compared the torsional resonance frequencies of the IL-coated substrates with those of bare substrates at the highest normal load applied during the friction measurements. If a direct contact of the probe with the substrate is established, the measured torsional resonance frequency on the IL-coated substrate must be the same as that on the bare substrate. However, as shown in Figs. 4 and 5 and Fig. S3, the torsional resonance frequencies were always higher on the IL-coated substrates than those on the bare substrates. This is indicative of completely different contact stiffnesses and substantiates the fact that the colloid probes and the substrates are not in a direct contact during the friction measurements.

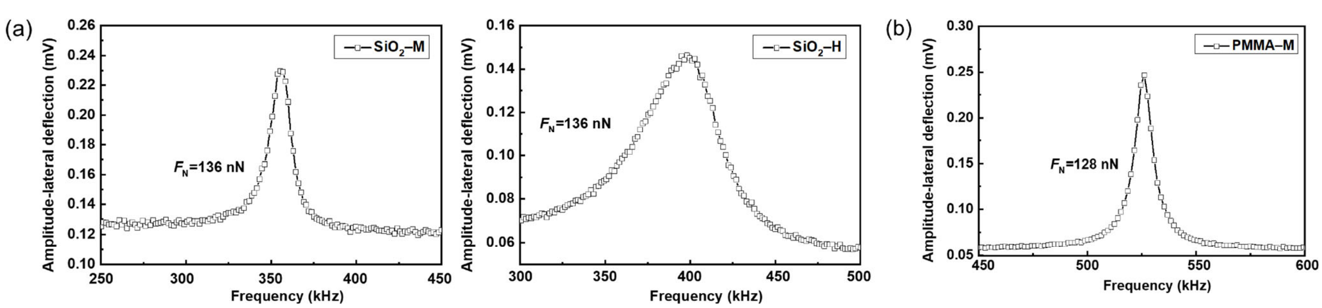


Fig. S3 Torsional resonance spectra from the lateral/friction force signal measured by the (a) SiO_2 and (b) PMMA colloid probes, at bare substrate mica (M) and HOPG (H) surfaces under higher normal loads (F_N) than the highest normal loads applied in the IL-substrate systems.

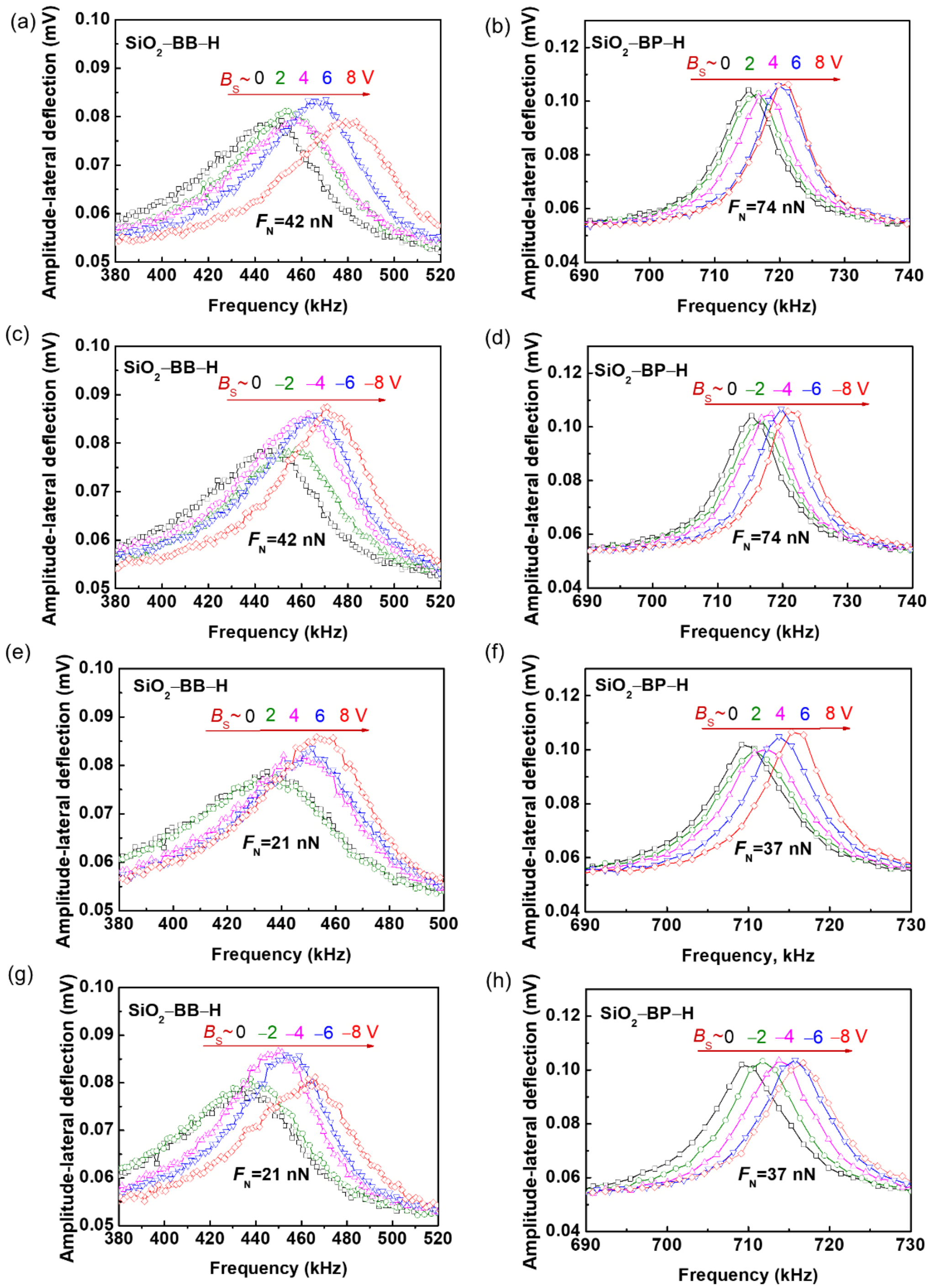


Fig. S4 (a–h) *In situ* changes of torsional resonance spectra recorded with the friction force obtained in Figs. 6(a) and 6(b) in the main text, under different sample bias (B_S), at normal load (F_N) of 42 and 21 nN for SiO_2–BB–H, and 74 and 37 nN for SiO_2–BP–H: (a, c, e, g) positive-bias and (b, d, f, h) negative-bias voltages.

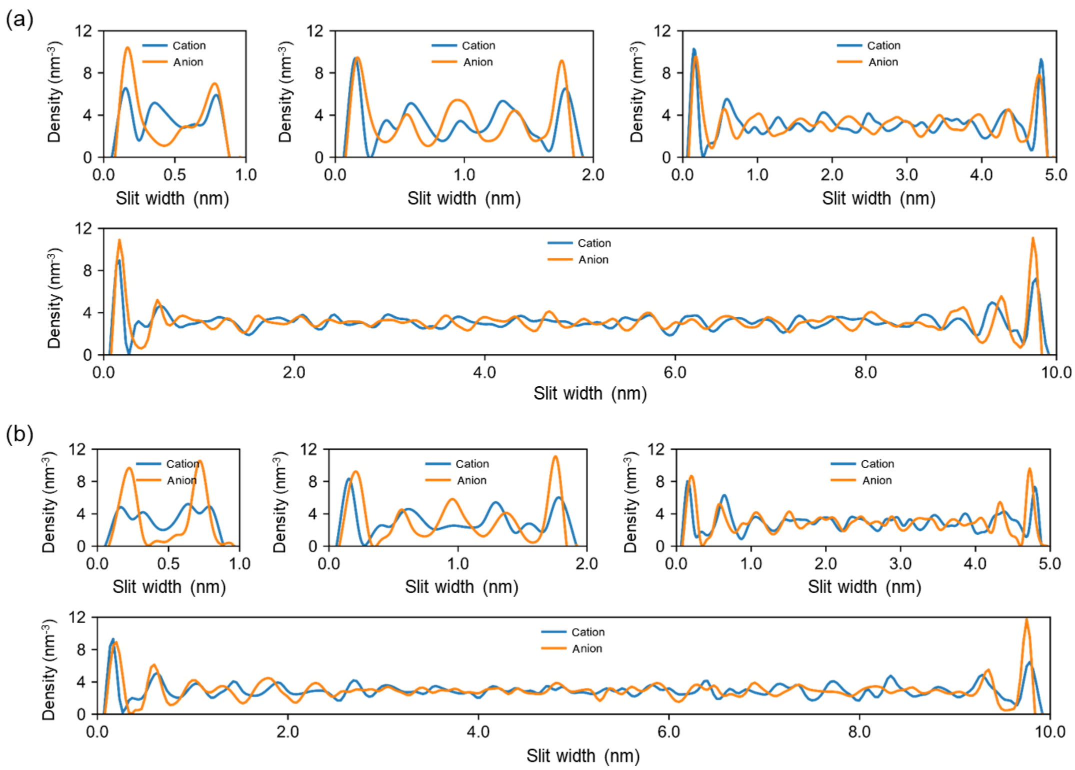


Fig. S5 Number density profiles of cations and anions of (a) BB and (b) BP, confined in uncharged grapheme–SiO_2 slit pores with width ranging from 1 to 10 nm. Top-left: 1 nm; top-middle: 2 nm; top-right: 5 nm; and bottom: 10 nm. Slit width = 0 nm represents three-layer graphene, and the other side of the slit pore represents SiO_2 surface.

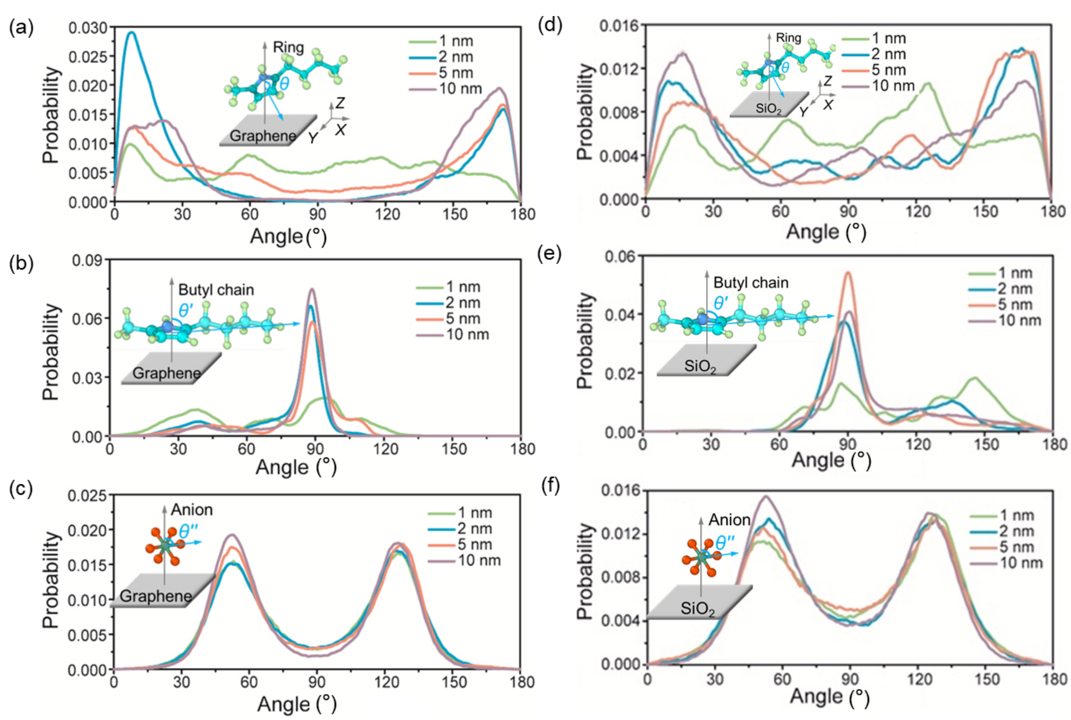


Fig. S6 Angular probability distributions for the imidazolium ring of cation $[BMIM]^{+}$, butyl chain, and anion $[PF_6]^{-}$ of BP IL on the surface of (a, b, c) graphene and (d, e, f) SiO_2 under confinement. The insets are typical configurations of the (a), (d) imidazolium ring; (b), (e) butyl chain; and (c), (f) anion on the surfaces.

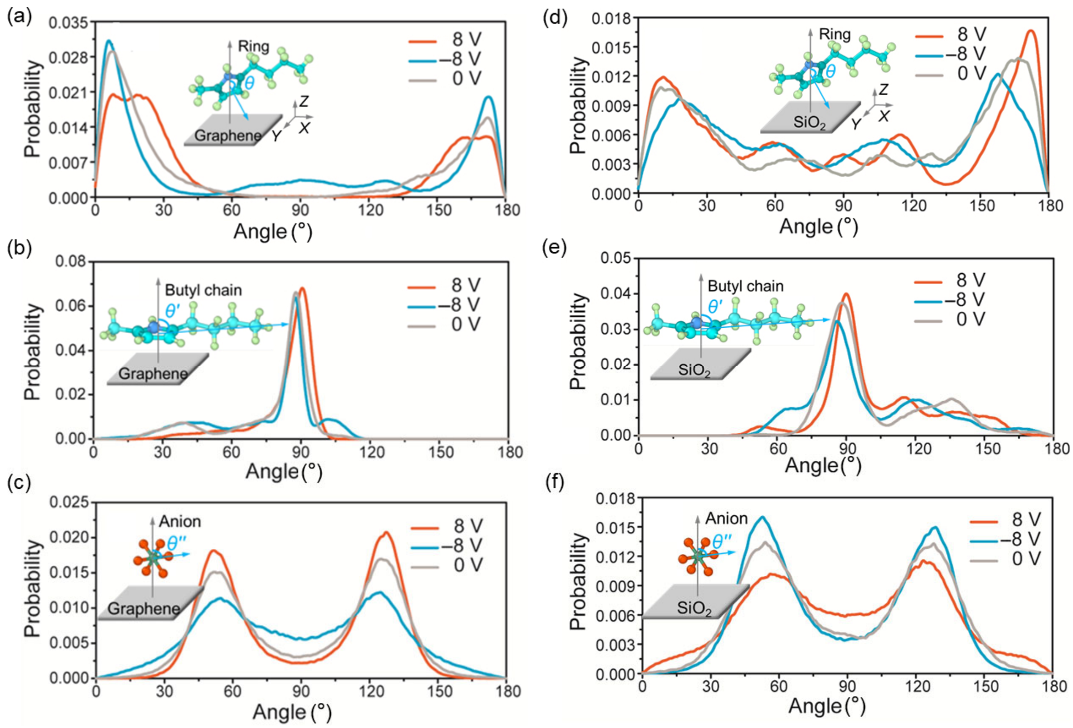


Fig. S7 Angular probability distributions for the imidazolium ring of cation $[BMIM]^+$, butyl chain, and anion $[PF_6]^-$ of BP on the surface of (a, b, c) graphene and (d, e, f) SiO_2, when IL is confined in a 2 nm slit pore with applied voltages of 0, −8, and 8 V.

Table S1 Number of ion pairs (*N*) of the ILs (BB and BP) confined in SiO_2–graphene slit with varying pore widths (1, 2, 5, and 10 nm).

Slit pore width (nm)	1	2	5	10
N (BB)	119	238	597	1,194
N (BP)	112	224	560	1,120

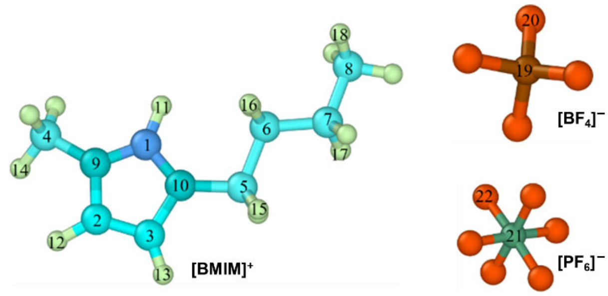
The Lennard–Jones 12–6 (Eq. (S1)) and Coulombic potential models (Eq. (S2)) are used to describe van der Waals (vdW), E_{vdW} and electrostatic interactions, E_{coul}, and the parameters (ε, σ, and q) are listed in Table S2:

$$E_{\mathrm{vdW}} = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] \tag{S1}$$

$$E_{\mathrm{coul}} = \frac{q_i \times q_j}{\varepsilon_{\mathrm{r}}\varepsilon_0 r} \tag{S2}$$

where ε and σ represent the parameters of size and energy well depth in the Lennard–Jones intermolecular pair potential energy, respectively, and r is the distance between two interacting atoms. q_i and q_j represent the charge of atom i and j, respectively; ε_{r} represents the relative dielectric constant of ILs; and ε_0 represents the dielectric constant of vacuum.

Table S2 Lennard–Jones parameters (ε, σ) and partial atomic charges (q) used in the ILs in this study.



ID	Atom type	ε (kcal/mol)	σ (Å)	q (e)
1	C	0.086	3.400	−0.006
2	C	0.086	3.400	−0.143
3	C	0.086	3.400	−0.218
4	C	0.109	3.400	−0.085
5	C	0.109	3.400	−0.015
6	C	0.109	3.400	0.011
7	C	0.109	3.400	0.031
8	C	0.109	3.400	−0.071
9	N	0.170	3.250	0.060
10	N	0.170	3.250	0.068
11	H	0.015	1.782	0.226
12	H	0.015	2.511	0.234
13	H	0.015	2.511	0.263
14	H	0.016	2.471	0.109
15	H	0.016	2.471	0.080
16	H	0.016	2.650	0.020
17	H	0.016	2.650	0.016
18	H	0.016	2.650	0.029
19	B	0.095	3.581	1.150
20	F	0.061	3.118	−0.538
21	P	0.200	3.742	0.756
22	F	0.061	3.118	−0.293
SiO_2	Si	0.310	3.804	0.122
	O	0.096	3.033	−0.061
Graphene	C	0.070	3.550	0.000

References

[S1] An R, Qiu X H, Shah F U, Riehemann K, Fuchs H. Controlling the nanoscale friction by layered ionic liquid films. *Phys Chem Chem Phys* **22**(26): 14941–14952 (2020)

[S2] Ally J, Vittorias E, Amirfazli A, Kappl M, Bonaccurso E, McNamee C E, Butt H-J. Interaction of a microsphere with a solid-supported liquid film. *Langmuir* **26**(14): 11797–11803 (2010)